

Qty. Description

1 NK 80-250/270 AA2F2AESBQQENW3



Note! Product picture may differ from actual product

Product No.: [98476530](#)

Non-self-priming, single-stage, centrifugal pump designed according to ISO 5199 with dimensions and rated performance according to EN 733. Flanges are PN 16 with dimensions according to EN 1092-2. The pump has an axial suction port, a radial discharge port and horizontal shaft. It is of the back pull-out design enabling removal of the motor, coupling, bearing bracket and impeller without disturbing the pump housing or pipework.

The unbalanced rubber bellows seal is according to DIN EN 12756.

The pump is fitted with a foot-mounted, fan-cooled asynchronous motor. Pump and motor are mounted on a common base frame.

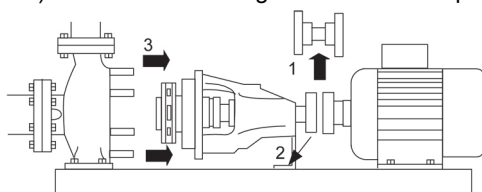
The product's minimum efficiency index (MEI) is greater or equal to 0.70. This is by the Commission Regulation (EU) considered as an indicative benchmark for best-performing water pump available on the market as from 1 January 2013.

Pump and motor are mounted on a common steel base frame in accordance with ISO 3661.

The back pull-out design together with a spacer coupling makes it possible to service the pump without dismantling the pump housing and motor from the base frame.

This saves realignment of pump and motor after service.

- 1) Remove coupling.
- 2) Remove the bolts in the bearing bracket support foot.
- 3) Remove the bearing bracket from the pump housing.



Pump

The pump is fitted with an unbalanced rubber bellows seal with torque transmission across the spring and around the bellows. Due to the bellows, the seal does not wear the shaft, and the axial movement is not prevented by deposits on the shaft.



Seal faces:

- Rotating seal ring material: silicon carbide (SiC)
- Stationary seat material: silicon carbide (SiC)

This material pairing is used where higher corrosion resistance is required. The high hardness of this material pairing offers good resistance against abrasive particles.

Secondary seal material: EPDM (ethylene-propylene rubber)

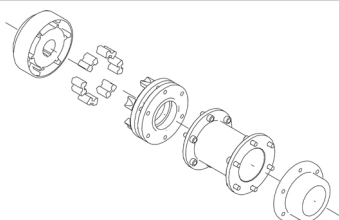
EPDM has excellent resistance to hot water. EPDM is not suitable for mineral oils.

The shaft is made of stainless steel and has a diameter of 32 mm where the coupling is mounted.

The pump uses a spacer coupling between the pump and motor shaft.

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The language on the pump nameplate is English.

Motor

The motor is a totally enclosed, fan-cooled motor with principal dimensions to IEC and DIN standards. Electrical tolerances comply with IEC 60034.

The motor efficiency is classified as IE3 in accordance with IEC 60034-30-1.

The motor has thermistors (PTC sensors) in the windings in accordance with DIN 44081/DIN 44082. The protection reacts to both slow- and quick-rising temperatures, e.g. constant overload and stalled conditions.

Thermal switches must be connected to an external control circuit in a way which ensures that the automatic reset cannot cause accidents. The motors must be connected to a motor-protective circuit breaker according to local regulations.

The motor can be connected to a variable speed drive for adjustment of pump performance to any duty point. Grundfos CUE offers a range of variable speed drives. Please find more information in Grundfos Product Center.

Further product details

Cast-iron parts have an epoxy-based coating made in a cathodic electro-deposition (CED) process. CED is a high-quality dip-painting process where an electrical field around the products ensures deposition of paint particles as a thin, well-controlled layer on the surface.

Technical data

Controls:

Frequency converter: NONE

Pressure sensor: N

Liquid:

Pumped liquid: Water

Liquid temperature range: -25 .. 120 °C

Selected liquid temperature: 20 °C

Density: 998.2 kg/m³

Technical:

Pump speed on which pump data are based: 1475 rpm

Rated flow: 120.3 m³/h

Pump with motor (Yes/No): Y

Rated head: 23.28 m

Actual impeller diameter: 270 mm

Nominal impeller diameter: 250

Code for shaft seal: BQQE

Mechanical seal type: Single

Curve tolerance: ISO9906:2012 3B

Bearing design: Standard

Materials:

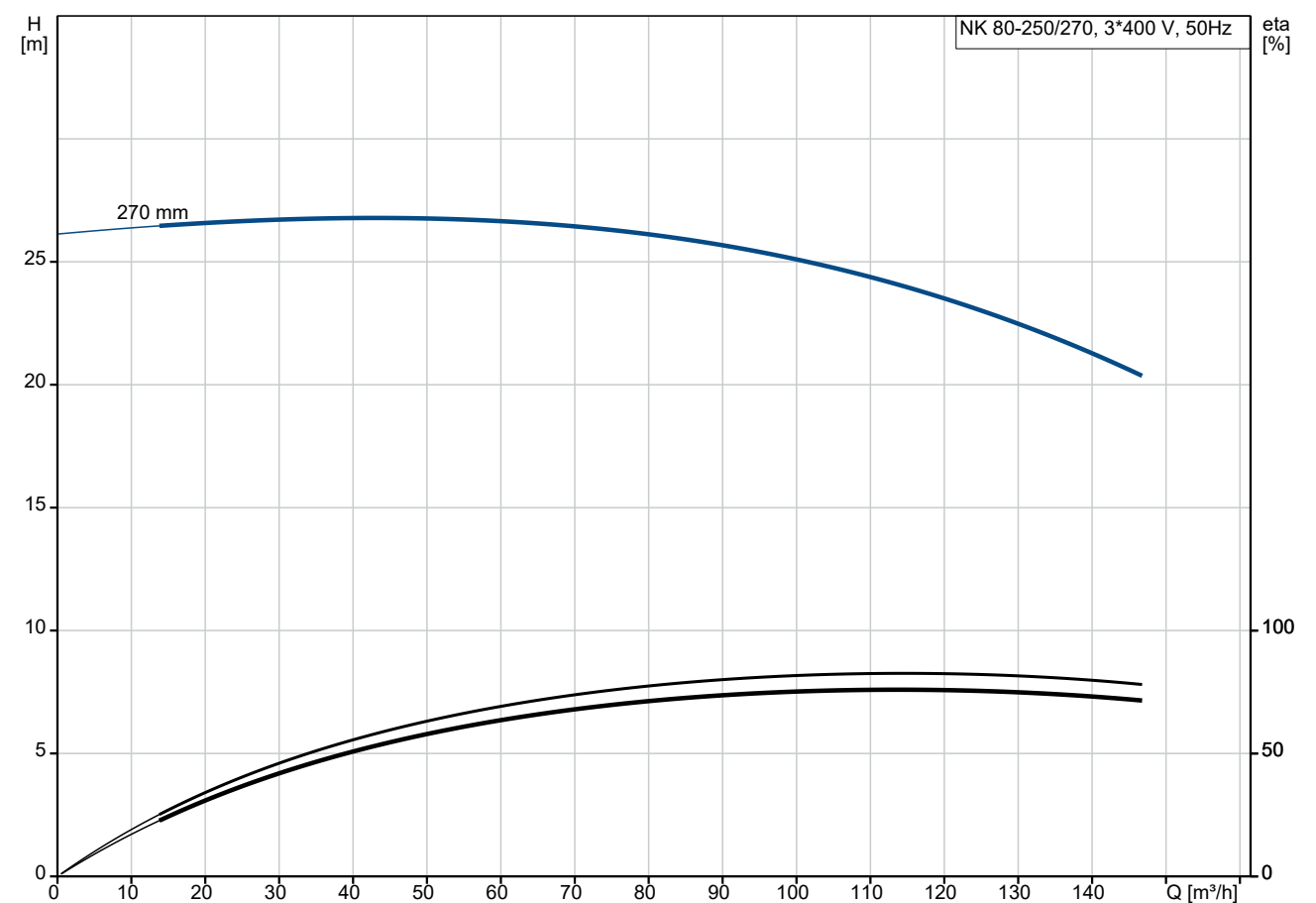
Pump housing: Cast iron
EN-GJL-250
ASTM class 35

Wear ring: Brass

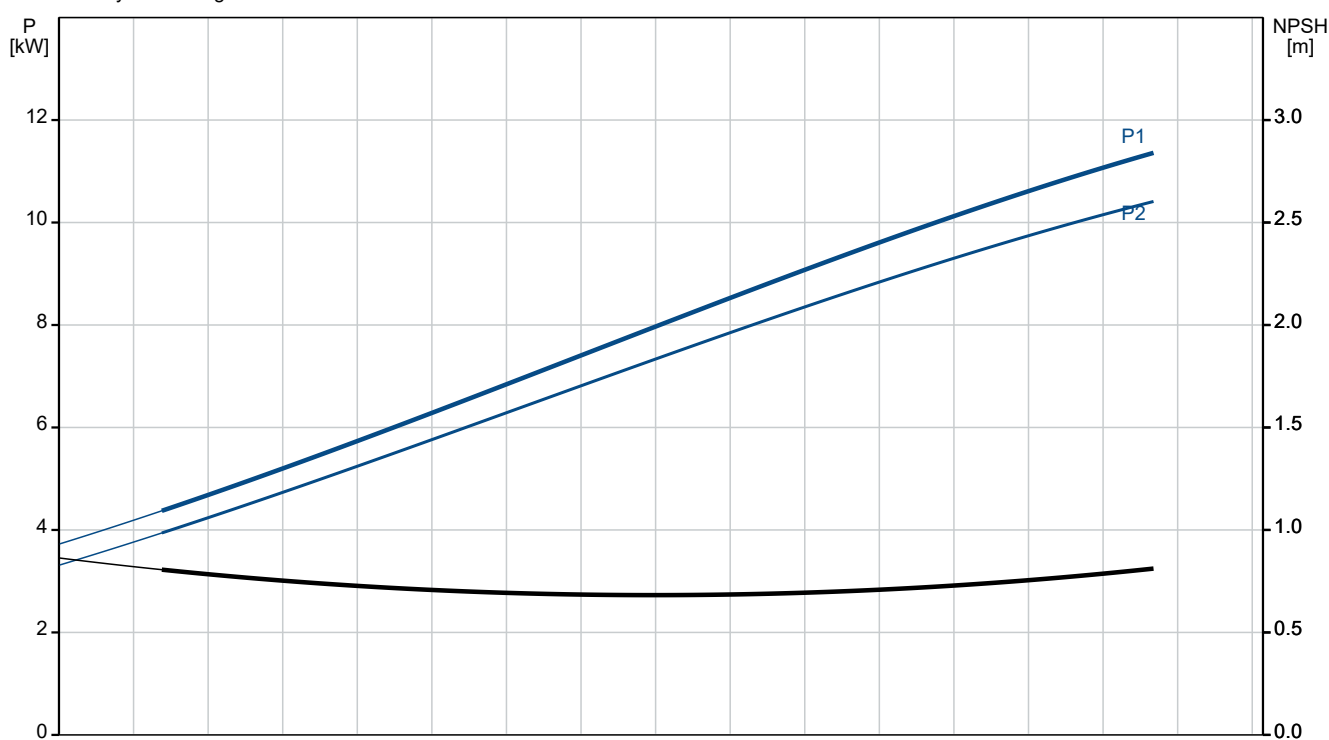
Qty. Description

1	Impeller: Cast iron EN-GJL-200 ASTM class 30 Internal pump house coating: CED Shaft: Stainless steel EN 1.4301 AISI 304 Installation: Maximum ambient temperature: 55 °C Maximum operating pressure: 16 bar Pipe connection standard: EN 1092-2 Type of inlet connection: DIN Type of outlet connection: DIN Size of inlet connection: DN 100 Size of outlet connection: DN 80 Pressure rating for connection: PN 16 Coupling type: Flexible w/spacer Base frame design: EN/ISO Code for base frame: 7 Grouting (Yes/No): N Electrical data: Motor type: SIEMENS Rated power - P2: 11 kW Mains frequency: 50 Hz Rated voltage: 3 x 380-420D/660-725Y V Rated current: 20.5/12 A Starting current: 680-680 % Cos phi - power factor: 0.84 Rated speed: 1475 rpm IE efficiency: IE3 91,4% IE Efficiency class: IE3 Motor efficiency at full load: 91.4-91.4 % Motor efficiency at 3/4 load: 91.9-91.9 % Motor efficiency at 1/2 load: 91.4-91.4 % Number of poles: 4 Enclosure class (IEC 34-5): IP55 Insulation class (IEC 85): F Motor No: 98957795 Bearing insulation type N-end: N Others: Minimum efficiency index, MEI ≥: 0.70 Net weight: 308 kg Gross weight: 335 kg Shipping volume: 0.717 m³ Language on pump nameplate: GB
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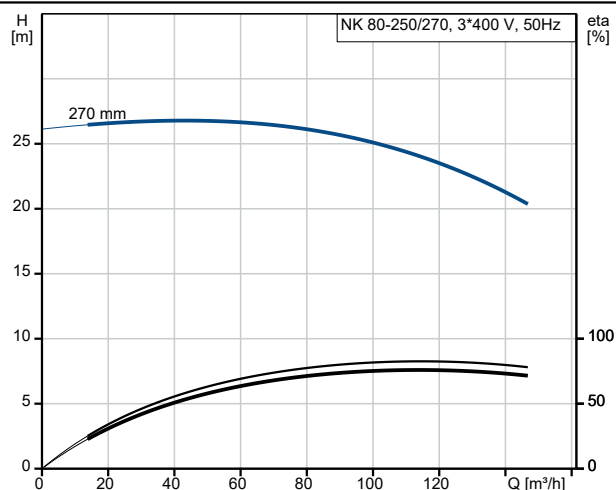
98476530 NK 80-250/270 AA2F2AESBQQENW3 50 Hz



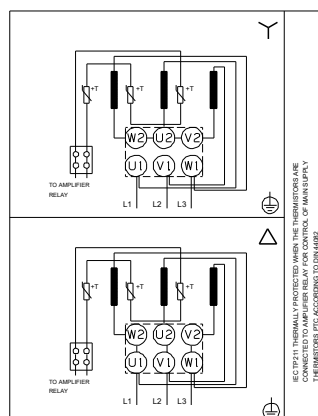
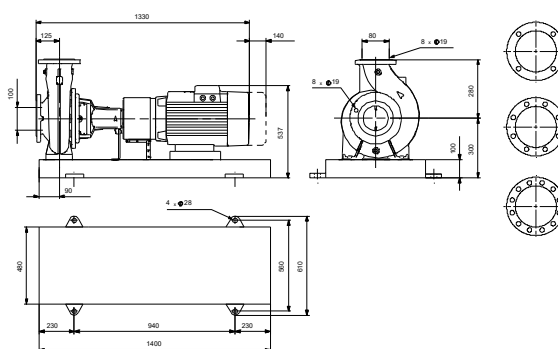
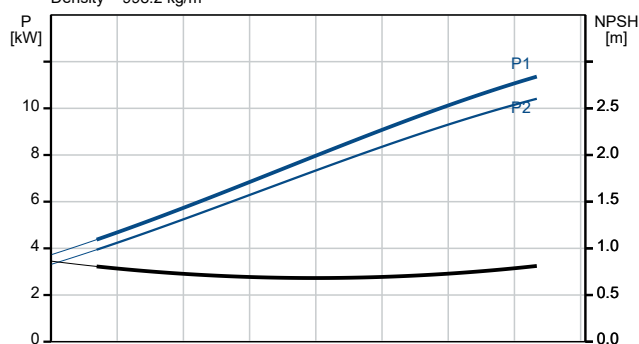
Pumped liquid = Water
Liquid temperature during operation = 20 °C
Density = 998.2 kg/m³



Description	Value
General information:	
Product name:	NK 80-250/270 AA2F2AESBQQENW3
Product No:	98476530
EAN number:	5711495735296
Technical:	
Pump speed on which pump data are based:	1475 rpm
Rated flow:	120.3 m³/h
Pump with motor (Yes/No):	Y
Rated head:	23.28 m
Actual impeller diameter:	270 mm
Nominal impeller diameter:	250
Shaft diameter:	32 mm
Code for shaft seal:	BQQE
Mechanical seal type:	Single
Curve tolerance:	ISO9906:2012 3B
Pump version:	A2
Bearing design:	Standard
Materials:	
Pump housing:	Cast iron
Pump housing:	EN-GJL-250
Pump housing:	ASTM class 35
Wear ring:	Brass
Impeller:	Cast iron
Impeller:	EN-GJL-200
Impeller:	ASTM class 30
Internal pump house coating:	CED
Material code:	A
Code for rubber:	E
Shaft:	Stainless steel
Shaft:	EN 1.4301
Shaft:	AISI 304
Installation:	
Maximum ambient temperature:	55 °C
Maximum operating pressure:	16 bar
Pipe connection standard:	EN 1092-2
Type of inlet connection:	DIN
Type of outlet connection:	DIN
Size of inlet connection:	DN 100
Size of outlet connection:	DN 80
Pressure rating for connection:	PN 16
Coupling type:	Flexible w/spacer
Base frame design:	EN/ISO
Code for base frame:	7
Grouting (Yes/No):	N
Connect code:	F
Liquid:	
Pumped liquid:	Water
Liquid temperature range:	-25 .. 120 °C
Selected liquid temperature:	20 °C
Density:	998.2 kg/m³
Electrical data:	
Motor type:	SIEMENS
Rated power - P2:	11 kW
Mains frequency:	50 Hz
Rated voltage:	3 x 380-420D/660-725Y V
Rated current:	20.5/12 A



Pumped liquid = Water
Liquid temperature during operation = 20 °C
Density = 998.2 kg/m³





Company name:

Created by:

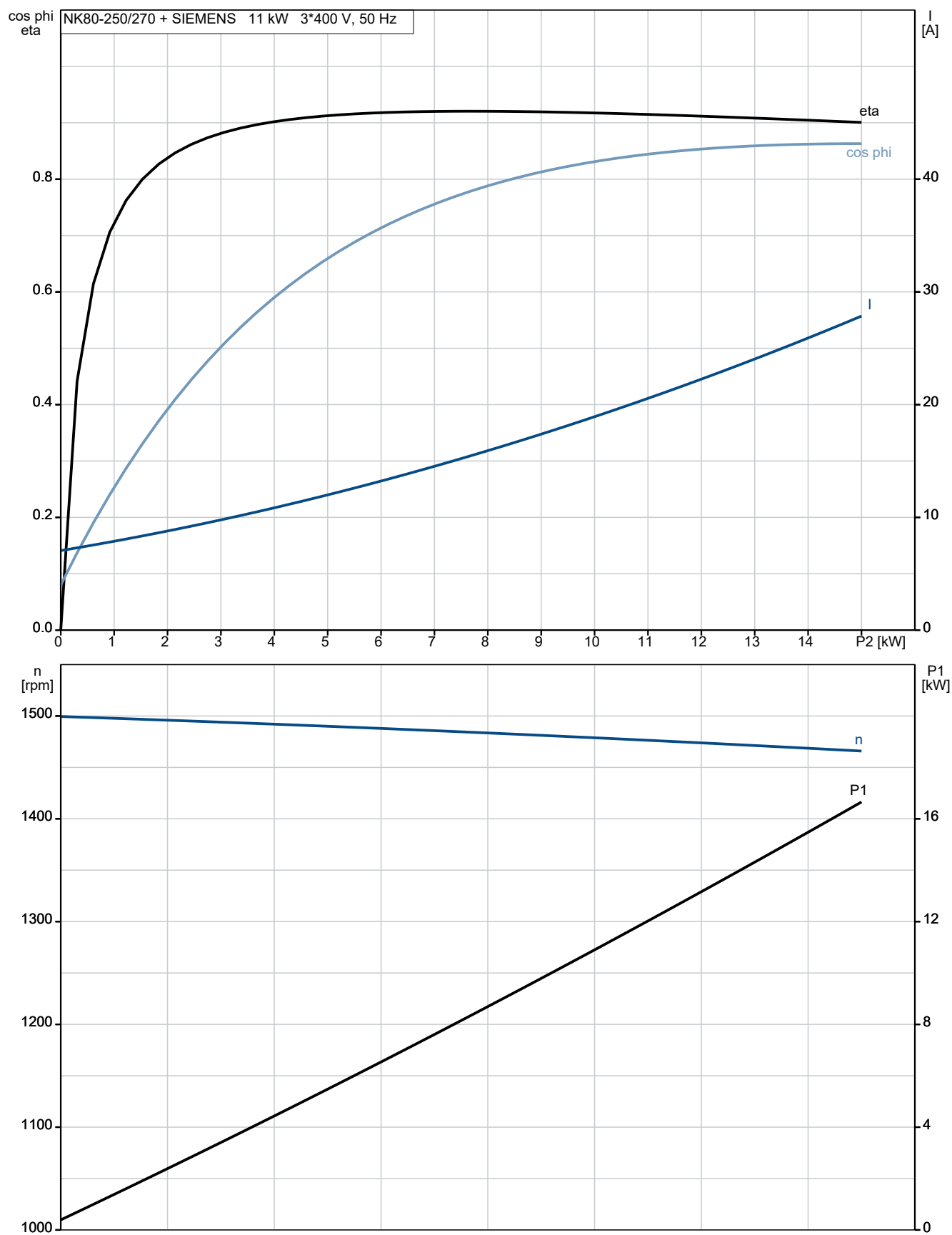
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Date:

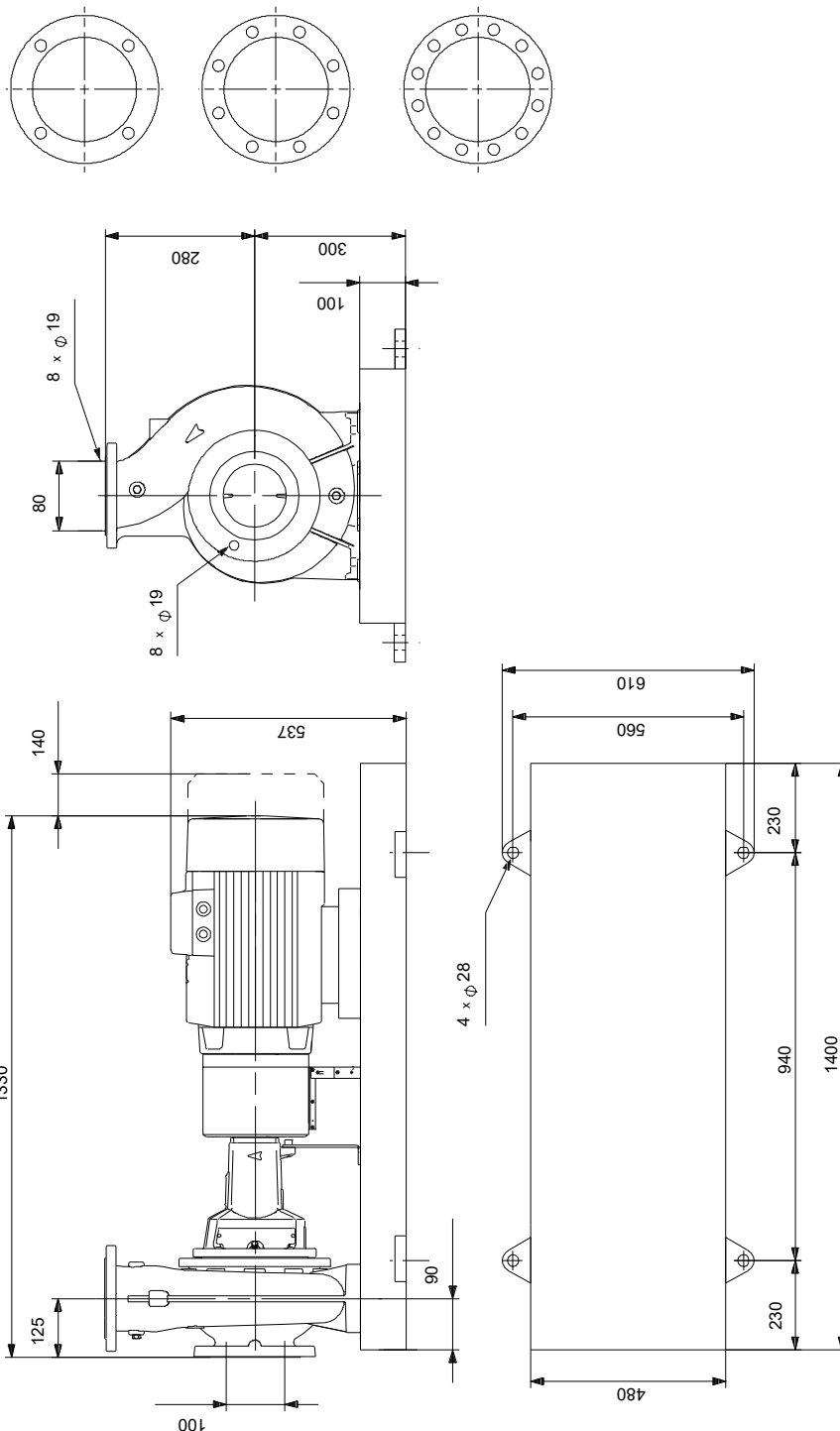
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Description	Value
Starting current:	680-680 %
Cos phi - power factor:	0.84
Rated speed:	1475 rpm
IE efficiency:	IE3 91,4%
IE Efficiency class:	IE3
Motor efficiency at full load:	91.4-91.4 %
Motor efficiency at 3/4 load:	91.9-91.9 %
Motor efficiency at 1/2 load:	91.4-91.4 %
Number of poles:	4
Enclosure class (IEC 34-5):	IP55
Insulation class (IEC 85):	F
Built-in motor protection:	PTC
Motor No:	98957795
Bearing insulation type N-end:	N
Controls:	
Frequency converter:	NONE
Pressure sensor:	N
Others:	
Minimum efficiency index, MEI ≥:	0.70
Net weight:	308 kg
Gross weight:	335 kg
Shipping volume:	0.717 m³
Language on pump nameplate:	GB

98476530 NK 80-250/270 AA2F2AESBQQENW3 50 Hz

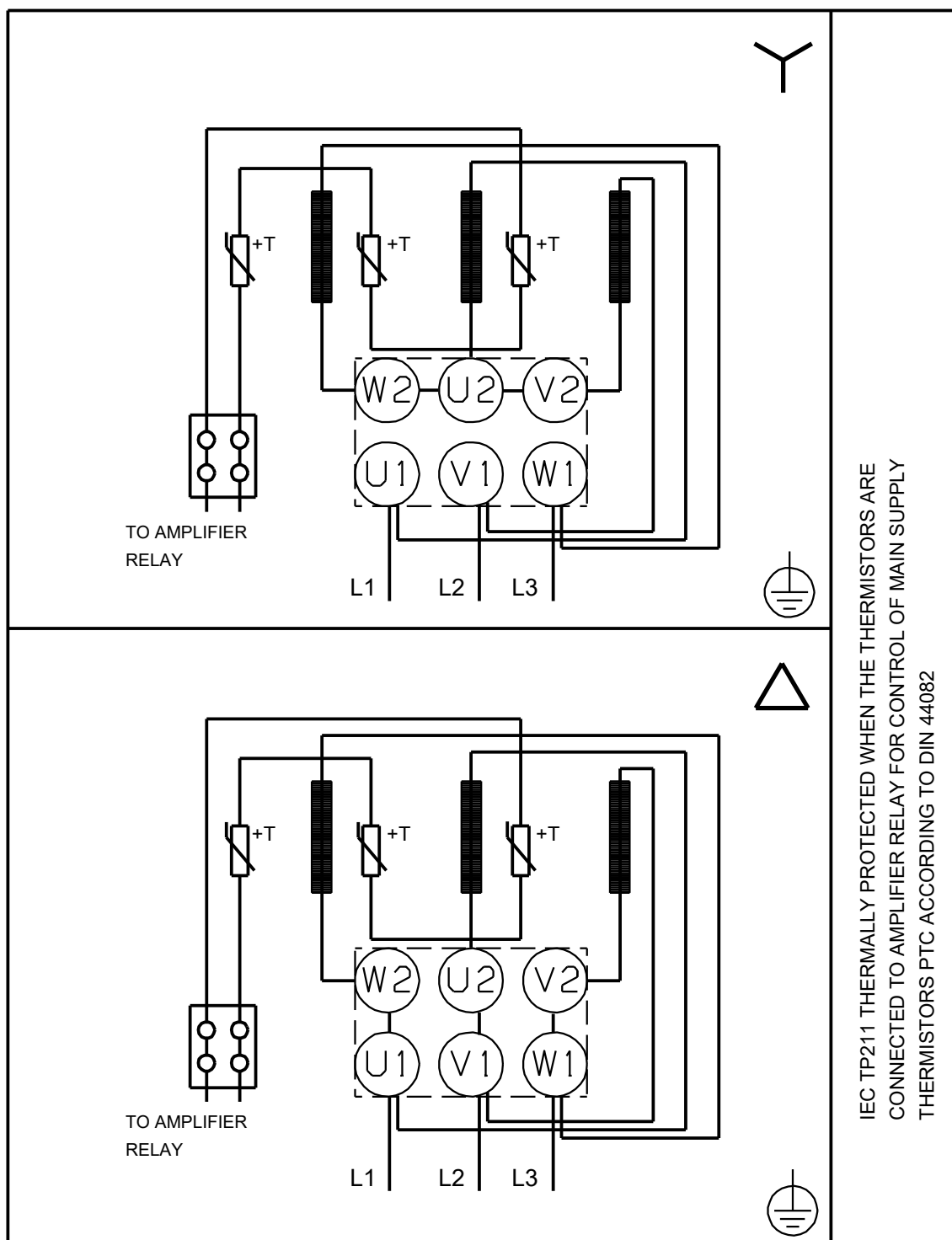


98476530 NK 80-250/270 AA2F2AESBQQENW3 50 Hz



Note! All units are in [mm] unless others are stated.
Disclaimer: This simplified dimensional drawing does not show all details.

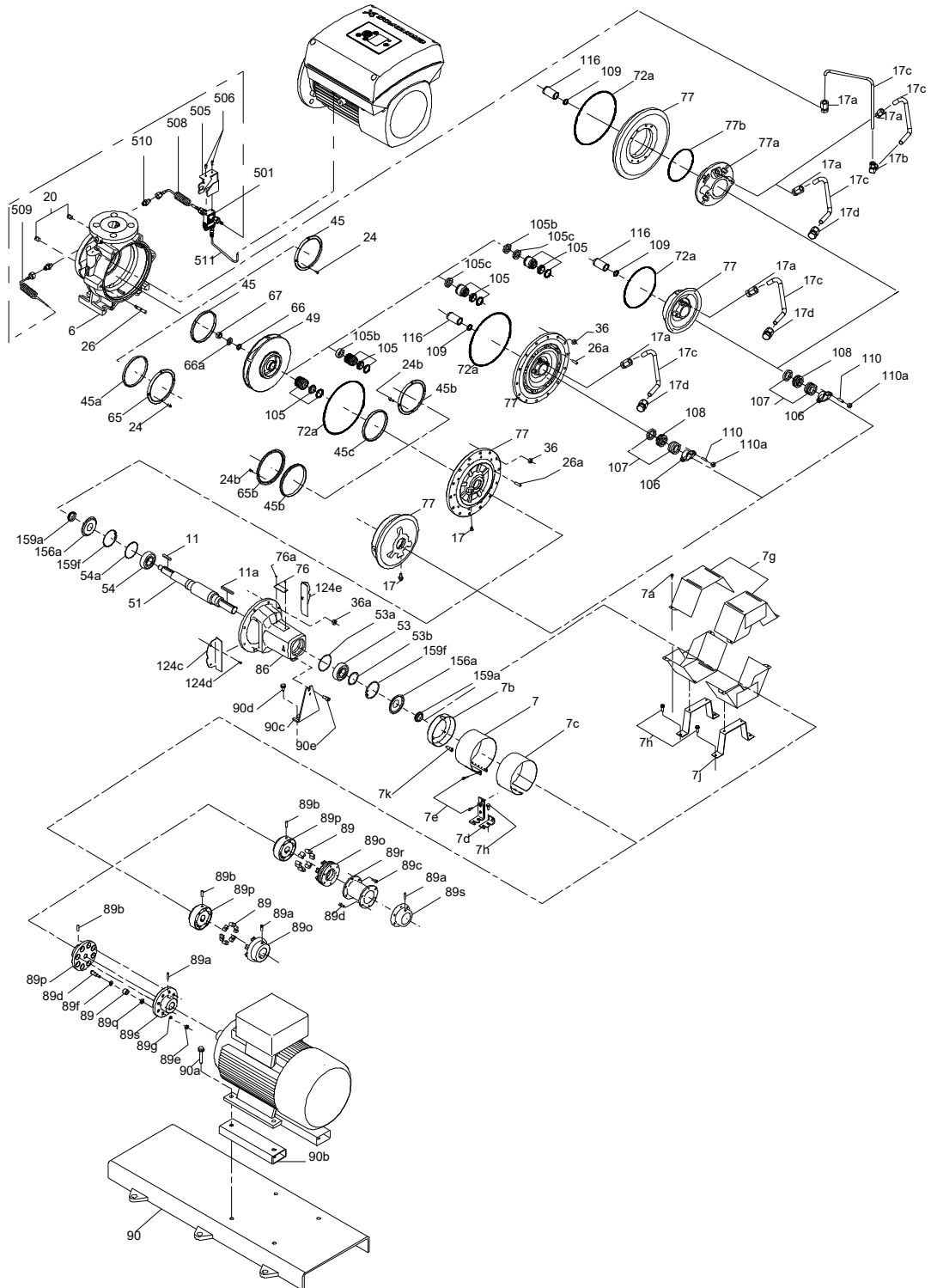
98476530 NK 80-250/270 AA2F2AESBQQENW3 50 Hz



IEC TP211 THERMALLY PROTECTED WHEN THE THERMISTORS ARE
CONNECTED TO AMPLIFIER RELAY FOR CONTROL OF MAIN SUPPLY
THERMISTORS PTC ACCORDING TO DIN 44082

Note! All units are in [mm] unless others are stated.

(NK)



Spare parts NK 80-250/270, Product No. 98476530

Produced After 2104 (production year and week number)

Pos	Description	Annotation	Classification Data	Part no.	Qty.	Unit
-	Kit, Bearing			96579599	1	pcs
53	Ball bearing				1	pcs
54a	O-ring		Diameter: 89,5		1	pcs
			Material type: EPDM			
			Thickness: 3			
54a	O-ring		Diameter: 89,5		1	pcs
			Material type: EPDM			
			Thickness: 3			
54	Ball bearing				1	pcs
156a	Bearing cover				2	pcs
159f	Retaining ring				2	pcs
159a	V-ring				2	pcs
-	Kit, Bearing bracket			95131794	1	pcs
7k	Hex socket head cap screw		Length (mm): 10		4	pcs
			Thread: M6			
36a	Nut		Thread: M10		6	pcs
76a	Rivet				2	pcs
86	Bearing bracket				1	pcs
124e	Shaft guard				1	pcs
124d	Pan head screw				4	pcs
124c	Shaft guard				1	pcs
-	Kit, Coupling			96579802	1	pcs
- 8	Coupling cpl.				1	pcs
89s	Spacer ring				1	pcs
89r	Spacer sleeve				1	pcs
89o	Coupling part spacer side				1	pcs
89b	Socket set screw		Length (mm): 8		1	pcs
			Thread: M6			
89.p	Coupling female				1	pcs
89.d	Hex socket head cap screw		Length (mm): 16		6	pcs
			Thread: M6			
89	Rubber blocks for coupling				6	pcs
11a	Parallel key				1	pcs
-	Kit, Coupling guard			98078801	1	pcs
7n	Retainer for coupling guard				1	pcs
7k	Hex socket head cap screw		Length (mm): 10		4	pcs
			Thread: M6			
- 7	Coupling guard cpl.				1	pcs
7e	Hex socket head cap screw				3	pcs
7c	Spacer f/coupling guard				1	pcs
7	Coupling guard cpl.				1	pcs
94	Lock nut		Thread: M6		2	pcs
-	Kit, Cover			95131790	1	pcs
18	Plug				1	pcs
36	Hex nut		Thread: M12		10	pcs
36	Nut		Thread: M10		8	pcs
77	Cover f/mech. seal				1	pcs
-	Kit, Foot for bearing bracket			96909170	1	pcs
36	Hex nut		Thread: M12		2	pcs
66	Washer				2	pcs
90e	Hex head screw		Length (mm): 25		1	pcs
			Thread: M12			
90d	Hex head bolt		Length (mm): 110		2	pcs

Pos	Description	Annotation	Classification Data	Part no.	Qty.	Unit
			Thread: M12			
90d	Hex head bolt		Length (mm): 90		2	pcs
			Thread: M12			
90d	Hex head bolt		Length (mm): 65		2	pcs
			Thread: M12			
90d	Hex head bolt		Length (mm): 55		2	pcs
			Thread: M12			
90d	Hex head screw		Length (mm): 35MM		2	pcs
			Thread: M12			
90d	Hex head screw		Length (mm): 40		2	pcs
			Thread: M12			
90d	Hex head screw		Length (mm): 30		2	pcs
			Thread: M12			
90c	Foot				1	pcs
-	Kit, Foot for coupling guard			96848478	1	pcs
7h	Hex head screw		Length (mm): 16		2	pcs
			Thread: M8			
-	7d	Foot cpl.			1	pcs
7l	Support f/coupling guard foot				1	pcs
7h	Hex socket head cap screw				2	pcs
7d	Foot for coupling guard				1	pcs
66	Washer				2	pcs
-	Kit, Nut			96939130	1	pcs
11	Parallel key				1	pcs
-	36	Nut			1	pcs
66a	Spring lock washer				1	pcs
66	Washer				1	pcs
67	Nut		Thread: M18		1	pcs
-	Kit, Pump housing			95131731	1	pcs
-	6	Pump housing			1	pcs
26	Staybolt		Length (mm): 40		10	pcs
			Thread: M12			
45	Wear ring				1	pcs
20	Plug				2	pcs
36	Hex nut		Thread: M12		10	pcs
-	Kit, Rubber blocks f. coupling			96480016	1	pcs
89	Rubber blocks for coupling				6	pcs
-	Kit, Shaft			96039655	1	pcs
11a	Parallel key				1	pcs
11	Parallel key				1	pcs
51	Shaft				1	pcs
66a	Spring lock washer				1	pcs
66	Washer				1	pcs
67	Nut		Thread: M18		1	pcs
-	Kit, Shaft cover			96809957	1	pcs
124e	Shaft guard				1	pcs
124d	Pan head screw				4	pcs
124c	Shaft guard				1	pcs
-	Spare, Wear ring			96797626	1	pcs
45	Wear ring				1	pcs
-	Spare, Wear ring			96797628	1	pcs
45	Wear ring				1	pcs
-	V-ring			96871238	1	pcs
159a	V-ring				2	pcs
-	Motor				1	pcs
156	Spare, Motor fan			98507837	1	pcs
151	Spare, Motor fan cover			98507838	1	pcs
176	Spare, Terminal board			98505488	1	pcs

Pos	Description	Annotation	Classification Data	Part no.	Qty.	Unit
164a	Spare, Terminal box			98505713	1	pcs
7k	Bulk, Hex socket head cap screw (20 PC)		Length (mm): 10 Thread: M6	97506946	4	pcs
28	Bulk, Hex head screw (10 PC)			92884780	4	pcs
28	Hex head screw		Length (mm): 25 Thread: M12	99522923	1	pcs
36	Bulk, Hex nut (20 PC)		Thread: M12	96620479	10	pcs
36	Bulk, Nut (20 PC)		Thread: M10	96620484	8	pcs
36	Nut		Thread: M10	92766626	8	pcs
49	Impeller			96591299	1	pcs
67	Bulk, Nut (5 PC)		Thread: M18	92526535	1	pcs
72a	Bulk, O-ring (5 PC)		Diameter: 278,99 Material type: EPDM Thickness: 3,53	92547634	1	pcs
72a	Bulk, O-ring (5 PC)		Diameter: 278,99 Material type: EPDM Thickness: 3,53	92526510	1	pcs
72a	Bulk, O-ring (10 PC)		Diameter: 278,99 Material type: EPDM Thickness: 3,53	97511839	1	pcs
72a	O-ring		Diameter: 278,99 Material type: EPDM Thickness: 3,53	97757673	1	pcs
77a	Bulk, Pan head screw (20 PC)			99266550	4	pcs
105b	Spacer			96591277	1	pcs
105	Bulk, Shaft seal (5 PC)			92514302	1	pcs
105	Shaft seal			98434906	1	pcs
109	Bulk, O-ring (10 PC)		Diameter: 89,5 Material type: EPDM Thickness: 3	97511809	2	pcs
154	Bulk, Ball bearing (5 PC)			98638919	2	pcs
156a	Bulk, Bearing cover (10 PC)			97506937	2	pcs